

MLOps on AWS SageMaker

Battery SoH / RUL Platform — Implementation Guide

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Audience: an engineer standing up the production MLOps stack for this project on AWS SageMaker.
Scope: how the existing local pipeline (src/*.py, models/<oem>/, the dashboards) maps onto SageMaker services — data → SoH target → features → training → **acceptance gate** → registry → scoring → monitoring — with the multi-OEM (Euler · Mahindra · Bajaj · Piaggio · Montra) fan-out baked in.

Read [README.md](#) first (the preprocessing/feature-generation runbook) and [SOH_RUL_TECHNICAL_REPORT.md](#) (the modelling detail). This document is the **operational** layer on top of them.

1 · TL;DR — the mapping

Pipeline stage (local today)	Reference code	SageMaker / AWS target
Raw telemetry in S3 (millions of tiny files)	battery-oem-data/parquet/<oem>/...	S3 data lake + a compaction job (Glue/EMR) — <i>do this first</i>
Extract cohort	src/import_cohort.py, src/montra_sample.py	SageMaker Processing (or Glue) job
SoH target (coulomb / BMS-capacity / reported)	src/soh.py,	same Processing job, per-OEM branch
Feature engineering → <oem>_featengg	src/euler_bms_soh.py	SageMaker Processing →
Train + backtest	src/features.py, src/*_features.py	Feature Store
	src/euler_train.py, src/oem_train.py	SageMaker Training job (script mode)
Evaluate (LOVO / held-out)	inside the trainers	Processing evaluation step → evaluation.json
Acceptance gate (coulomb yardstick)	src/euler_accept_gate.py (exit code = verdict)	Pipeline ConditionStep gating model approval
Model registry (registry.json/diagnostics.json + latest.pkl)	models/<oem>/	SageMaker Model Registry (one Model Package Group per OEM)

Pipeline stage (local today)	Reference code	SageMaker / AWS target
Deploy = load_latest(oem) in dashboards	oem_train.load_latest	Batch Transform (monthly) + optional Serverless Endpoint SageMaker Pipelines , parameterised by oem
Orchestration (run scripts by hand)	—	EventBridge Scheduler → Pipeline
Schedule (quarterly retrain, per docstring)	scripts/retrain_*.sh (implied)	SageMaker Model Monitor + CloudWatch
Drift / accuracy watch (none yet)	—	IAM roles + Secrets Manager (no long-lived keys)
Secrets (.env, Redshift creds)	gitignored	

The single most important architectural decision is **compaction of the tiny-files layout** (§3). Everything downstream is standard SageMaker once the lake is query-shaped.

2 · Target architecture

(Diagram — see the Mermaid block in the Markdown source docs/MLOPS_SAGEMAKER.md, which renders on GitHub.)

Cadence (from euler_train.py): **monthly** feature refresh + batch scoring; **quarterly** full retrain (as more vehicles age toward the 80 %/70 % EoL line, the training signal strengthens). Both are the same Pipeline invoked with different parameters.

3 · The data lake and the tiny-files problem (do this first)

The raw feed is day-partitioned parquet with **~2 VINs per file** — Piaggio ~308k files, Montra ~200k+, the Mahindra native feed on the order of 70M. A SageMaker Training or Processing job that reads these directly will spend all its time on S3 LIST/GET, not compute. Fix it **upstream, once**:

1. **Compaction job** — AWS **Glue (Spark)** or **EMR Serverless** reads the raw partitions and rewrites to a **curated zone** partitioned by oem + month, target file size ~128-256 MB, vin as a dictionary column. Schedule it to append the latest ingest partitions daily. This is the one step that genuinely needs Spark; the rest is comfortable in pandas on a single Processing instance.
2. **Glue Data Catalog + Athena** over the curated zone, so the extract step is a partition-pruned query rather than a directory crawl.
3. Keep the **sentinel clipping** (SoC 0-100, $|I| \leq 400$ A, V 20-120, resCapacity 1-500) in the compaction step so garbage never reaches feature code — same bounds as [README.md §2](#).

If you skip compaction, at minimum use **S3DataDistributionType=ShardedByS3Key** and threaded batched reads in the Processing job — but compaction is strictly better and removes the RAM-pressure of vin-category casts.

4 · Component detail

4.1 Preprocessing + SoH + features → SageMaker Processing

One **ProcessingStep** per OEM produces the <oem>_featengg table (the 32-column schema in [README.md §5](#)). It wraps the *existing* code with almost no change:

- Container: a custom image (§6) with pandas · pyarrow · numpy · scikit-learn · scipy.

- Entry point: a thin processing/build_features.py that calls the OEM's *_features.py template (piaggio_features.py for coulomb OEMs, montra_features.py/euler_bms_soh.py for BMS-capacity OEMs, the bajaj_model path for reported-SoH).
- **SoH method branches on the feed**, exactly as config.SOH_METHOD already encodes:

Feed carries	Method	Code
signed current + SoC (Mahindra, Piaggio via intellicar)	coulomb counting → isotonic	soh.coulomb_capacity_monthly → capacity_to_soh
remaining-capacity (Euler, Montra)	BMS remaining-capacity → normalise → isotonic	euler_bms_soh.py, montra_features.py
reported SoH only (Bajaj)	monthly median, kept non-increasing	bajaj_model

- Inputs: curated S3 zone (or Athena unload). Outputs: s3://.../featengg/<oem>/ **and** an ingest into Feature Store (§4.2). Instance: ml.m5.2xlarge/ml.r5 for the coulomb OEMs (memory-bound pooling), scale by cohort.

src/mahindra_soh_methods.py and dashboard/mahindra_sources.py are **local-only, un-committed** helpers (the AWS build must not depend on them being in git). Port the logic into the Processing entry point, not the other way round.

4.2 Feature Store — the featengg contract

The featengg table is *already* a clean feature-group shape: **one row per (vin, month)**, immutable once a month closes. Model it as a **SageMaker Feature Group**:

- **Record identifier**: vin; **event time**: ymd (month timestamp).
- **Offline store** (S3 + Glue catalog) → training and batch scoring read point-in-time-correct history.
- **Online store** → the customer app / real-time endpoint reads the latest vin-month at low latency.
- **Do not rename or drop columns** — src/model.py, euler_model, bajaj_model, oem_train.py, and every dashboard read the schema positionally-by-name; missing per-OEM columns (dte_mean, cell imbalance) stay NaN and the tree models tolerate it.

One feature group per OEM keeps the per-OEM column availability explicit and lets fleets version independently.

4.3 Training → SageMaker Training job (script mode)

euler_train.py and oem_train.py are already self-contained trainers that emit a versioned bundle + registry + diagnostics. Adapt them to the SageMaker Training contract:

- **Read** features from the Feature Store offline store / SM_CHANNEL_TRAIN instead of data/redshift/...
- **Write** the model bundle to SM_MODEL_DIR (becomes model.tar.gz) instead of models/<oem>/latest.pkl.
- **Emit metrics** as a evaluation.json PropertyFile and via print() regex metric definitions so they land in the Training job's CloudWatch metrics.
- Keep the bundle shape unchanged: Euler {rate_model, traj_model, band, meta}; the generic OEMs {model, meta} where model = mod.train_quantiles(mod.build_transitions(m)).

Instance: CPU is enough (ml.m5.xlarge) — the models are LightGBM/XGBoost + scikit-learn isotonic. The torch/gpytorch/chronos GPU stack in requirements.txt is **experimental/optional**; keep it out of the production training image unless you deploy those variants (then ml.g5).

Per-OEM config currently lives in oem_train.CFG (module, eol, warr_yr). Promote it to **Pipeline parameters** so one pipeline definition serves every fleet (§5).

4.4 Evaluation / backtest

The trainers already run an honest generalisation check — **LOVO** (leave-one-vehicle-out) for Euler, a by-vehicle **60/20/20** held-out backtest for the others, each reporting `overall_rmse`, `degrading_rmse`, and a `persistence_rmse` baseline (see `models/euler/diagnostics.json`). Split this into its own **ProcessingStep** so the metrics are a first-class `PropertyFile` the gate and the registry can both consume. The degrading-cohort-vs-persistence comparison is the number that decides whether a fleet's model is worth deploying (Bajaj beats persistence; the noisy low-util Mahindra/Piaggio fleets don't — that fact should be visible in the registry, not buried).

4.5 The acceptance gate → `ConditionStep` (the crown jewel)

`src/euler_accept_gate.py` is the project's standing safeguard and it maps **directly** onto SageMaker MLOps:

- It scores a **candidate** SoH target's forecasts against a **physically independent** yardstick (coulomb full-charge SoH), on the coulomb-confirmed **decliner cohort**, and returns PASS/FAIL with a **nonzero exit code on FAIL** — purpose-built to gate a promotion script.
- In the Pipeline: run it as a **ProcessingStep** that writes `gate.json` (`{"verdict": "PASS"|"FAIL", ...}`), then a **ConditionStep** (`ConditionEquals(verdict, "PASS")`) whose *then*-branch is `RegisterModel(..., ModelApprovalStatus="Approved")` and whose *else*-branch registers as `PendingManualApproval` and fires an SNS alert. **Never auto-approve a target change that regresses the decliner cohort** — that is the whole point of the gate (the first `soh_label` retrain looked +30 % better against a *non-independent* yardstick and was correctly rejected; the hybrid target passed and shipped as `euler_20260707`).
- The gate needs the independent coulomb yardstick (`euler_full_charge_soh.py` output). Where a fleet has no independent yardstick (Bajaj has no current at all; Piaggio's would require scanning 308k intellicar files), the gate is **not applicable** — those OEMs keep their already-isotonic target and skip the `ConditionStep`. Encode that as a per-OEM `has_gate` pipeline parameter.

```
from sagemaker.workflow.condition_step import ConditionStep
from sagemaker.workflow.conditions import ConditionEquals
from sagemaker.workflow.functions import JsonGet

gate_pass = ConditionEquals(
    left=JsonGet(step_name=gate_step.name, property_file=gate_report, json_path="verdict"),
    right="PASS")

ConditionStep(
    name="AcceptanceGate",
    conditions=[gate_pass],
    if_steps=[register_approved],          # ModelApprovalStatus="Approved"
    else_steps=[register_pending, notify], # PendingManualApproval + SNS
)
```

4.6 Model Registry

Replace the hand-maintained `registry.json` / `diagnostics.json` with the **SageMaker Model Registry**:

- **One Model Package Group per OEM** (`soh-forecaster-euler`, `-mahindra`, ...).
- Each training run → a **Model Package** carrying the `model.tar.gz`, the evaluation metrics (`overall_rmse`/`degrading_rmse`/`persistence_rmse`/`band_coverage`), the acceptance-gate verdict, and the training lineage (which featengg snapshot, which git SHA).
- **Approval status is the deployment control**: batch scoring and endpoints deploy the *latest Approved* package — the registry equivalent of today's `latest.pkl`. `load_latest(oem)` becomes "fetch latest Approved Model Package for the group".
- Keep the per-run history you get from `registry.json` today — the Model Registry gives it to you for free, plus lineage back to the exact data.

4.7 Scoring / serving

The product is **monthly, fleet-wide** (per-vehicle SoH forecast → warranty-risk tier → RUL-in-km → dashboards), so the default is **batch, not real-time**:

- **Batch Transform** (or a scheduled Scoring ProcessingStep) over the current-month feature snapshot → write forecasts + P10/P90 bands + warranty-risk tier to S3 and back to Redshift (soh_etl.*) for the dashboards.
- **Optional Serverless / real-time endpoint** only for the **customer app** (dashboard/battery_status.py) if it needs on-demand per-VIN answers; it reads the online Feature Store for the vehicle's latest month and returns the forecast. Serverless keeps cost near zero at low, spiky traffic.
- The current Streamlit dashboards can stay as-is initially, repointed from load_latest()/local parquet to the batch-scored S3/Redshift outputs.

4.8 Orchestration — SageMaker Pipelines

One Pipeline definition, **ParameterString("oem")** + the per-OEM knobs (module, eol, warr_yr, has_gate, soh_method). Steps: Process(features) → FeatureStoreIngest → Train → Process(backtest) → Process(gate) → ConditionStep → RegisterModel. Trigger once per OEM (a thin loop or a Step Functions Map), so onboarding a new OEM is “*add a parameter set*”, not new pipeline code — the same scaling property oem_train.CFG gives you today. The concrete per-OEM parameter sets are in §10.

4.9 Monitoring — Model Monitor + CloudWatch

Two monitors, plus one project-specific nuance:

- **Data-quality / drift monitor** on the feature inputs — baseline the featengg distributions and alarm when km_month, ah_throughput, temp_mean, SoC habits, or the sentinel-clip rejection rate drift. This catches a feed schema change or a new OEM firmware revision before it silently corrupts SoH.
- **Model-quality monitor** — as real SoH is observed in later months, join predictions to actuals and track realized forecast RMSE / bias per OEM in CloudWatch; alarm on regression. **Ground truth arrives with a months-long lag** (you only learn a forecast was wrong once the battery has aged), so this monitor is a slow trailing signal — design the alarm windows accordingly, and lean on the acceptance gate + backtest for *pre-deployment* confidence.
- **Cold-start fleets** (Montra, young Bajaj) sit ~flat at 100 % with zero decliners — their models are honest **placeholders**. Tag them in the registry and **suppress “no drift / perfect fit” alarms**; the signal only becomes real once the fleet ages. Likewise remember the **aged-vehicle scarcity** bias (the only EoL examples are early failures) makes at-risk % an **upper bound** — surface it as such in monitoring dashboards.

4.10 Scheduling & retraining

EventBridge Scheduler → Pipeline: monthly (feature refresh + batch score) and quarterly (full retrain, all OEMs). The retrain writes new Model Packages; the gate + backtest decide whether they auto-approve. Keep the “does this retrain actually improve accuracy?” question answerable — the registry's per-run metric history is how you avoid churning a model that isn't getting better.

4.11 CI/CD

- **Code** stays in git (this repo). A **CodePipeline / GitHub Actions** build produces the Processing/Training **container image** (ECR) from requirements.txt and runs a smoke test, then updates the SageMaker Pipeline definition.
- Consider a **SageMaker Project** MLOps template to wire repo → build → pipeline → registry → deploy with the approval workflow pre-built.

- registry.json/diagnostics.json are tracked in git today as a poor-man's registry; once the Model Registry is authoritative they become redundant (keep emitting them only if a dashboard still reads them).

5 • Multi-OEM parameterisation (the scaling pattern)

Everything OEM-specific is already centralised — mirror it as Pipeline parameters, not forked pipelines:

Knob	Source today	Pipeline parameter
SoH method	config.SOH_METHOD	soh_method (coulomb / bms_capacity / reported)
EoL threshold	oem_train.CFG[oem]["eol"] (80, Bajaj 70)	eol_pct
Warranty term	config.FLEET_WARRANTY / WARRANTY	warr_years, warr_km
Model module	CFG["module"] (model / bajaj_model / euler_model)	model_module
Independent yardstick exists?	(Euler only)	has_gate

Adding an OEM = audit its feed → pick soh_method → add a parameter set → run the pipeline. The [“Add a new OEM”](#) section of the README is the human checklist; this is its automation.

6 • Security, IAM, networking

- **No long-lived credentials.** The local .env (AWS keys, S3_BUCKET=oem-data-iot, ap-south-1) and the scratchpad redshift.env do **not** move to the cloud — replace with **IAM execution roles** (least-privilege per step: Processing role reads raw+curated S3, Training role reads Feature Store, scoring role writes forecasts) and **Secrets Manager** for the Redshift connection (used read-only, readonly=True).
- **Never commit secrets;** keep .env gitignored. This document and the Pipeline definitions must reference secrets by ARN, never by value.
- **Encryption:** KMS on the S3 zones, Feature Store, and model artifacts; **VPC** endpoints for S3 / SageMaker so telemetry never traverses the public internet; scope the SageMaker roles to the oem-data-iot bucket and the specific Feature Groups.
- **Region:** everything in ap-south-1 alongside the data to avoid egress and latency.

7 • Cost & scaling notes

- **Compaction (\$3) is the biggest lever** — it turns 308k-file scans into partition-pruned reads and cuts Processing wall-clock (and cost) by an order of magnitude.
- **Training is cheap** (CPU LightGBM/XGBoost, minutes) — use **Spot** for training and backtest steps.
- **Batch Transform monthly** beats an always-on endpoint for a fleet-batch product; only add a Serverless endpoint for genuine on-demand customer traffic.
- **Right-size Processing** by cohort: coulomb OEMs (pooled (vin,month) integration) are memory-bound — prefer ml.r5; BMS-capacity OEMs are lighter.

8 · Phased migration (crawl → walk → run)

1. **Crawl — lift the batch onto SageMaker.** Compaction job (§3) + one Processing job that runs the existing `*_features.py` for one OEM (Euler) → `featengg` in S3. Train with `euler_train.py` in a Training job. Manual approval, manual batch score. *Value: reproducible, roled, no laptop.*
2. **Walk — Feature Store + Pipeline + Registry.** Move `featengg` into Feature Store; wrap `Process→Train→Backtest` in a Pipeline; register to the Model Registry; wire `load_latest` → `latest-Approved`. Parameterise by OEM and onboard Mahindra/Bajaj/Piaggio/Montra.
3. **Run — gate + monitoring + schedule.** Add the acceptance-gate `ConditionStep` (Euler), Model Monitor (data
 - model quality), EventBridge monthly/quarterly triggers, and CI/CD for the container + pipeline. Customer-app endpoint if/when on-demand traffic warrants it.

9 · Project-specific gotchas to carry into the design

- **Tiny files dominate everything** — compact first (§3).
- **SoH method is per-feed, not global** — the branch in §4.1 is load-bearing; a wrong method silently produces a plausible-but-wrong target.
- **The yardstick-independence trap** — a smoother SoH target lowers *self*-error without being more accurate; the gate scores against a *physically independent* signal for exactly this reason. Do not let a metric that shares provenance with the target become the promotion criterion.
- **Ground truth lags** — model-quality monitoring is a trailing signal; pre-deployment backtest + gate carry the confidence.
- **Cold-start fleets are placeholders** — don't over-trust (or alarm on) a flat new-fleet model.
- **At-risk % is an upper bound** — aged-vehicle scarcity biases the training signal toward early failures; communicate the forecast with its P10/P90 band, never as a point estimate.

10 · Per-OEM implementation flows

One parameterised pipeline, **five parameter sets**. The steps are identical for every OEM; what changes is the *feed audit* → *SoH method* → *whether the acceptance gate applies* → *whether the fleet is mature enough to deploy a real model*. The audit decides everything downstream:

(Diagram — see the Mermaid block in the Markdown source `docs/MLOPS_SAGEMAKER.md`, which renders on GitHub.)

10.0 · Parameter sets & deltas

OEM	Extraction	SoH method	Model module (EoL / warranty)	Gate	Fleet	Dominant gotcha
Euler	dense batch	BMS-capacity → recovery-aware clean → hybrid	<code>euler_model</code> (80 % / 3 y-80k)	YES	mature (120 veh, 74 degraders)	SoH artifacts → <i>needs</i> the clean+gate
Mahindra	two-feed: native (~70M tiny files) + intellicar (signed I)	coulomb (intellicar); native-only → Bayesian behaviour SoH	<code>model</code> (80 % / 3 y-120k)	no	large, ~98 % native-only	70M tiny files; odometer logged late

OEM	Extraction	SoH method	Model module (EoL / warranty)	Gate	Fleet	Dominant gotcha
Bajaj	dense native	reported BMS SoH, non-increasing	bajaj_model (70 % / 5 y·120k)	no	young (~10 mo)	no I/V → cycles/temp/eff features; odo in metres
Piaggio	intellicar (SoH) + native (features)	coulomb (intellicar)	model (80 % / 3 y·100k)	no	noisy low-util	308k tiny intellicar files; native V 100 % null
Montra	day-sampled 10-veh POC	BMS-capacity (current unsigned)	model (80 % / 3 y·100k)	no	new POC — place-holder	unsigned current → no coulomb; flat fleet

10.1 · Euler — the full path (the only OEM that activates the gate)

Extract (dense batch) → **two** SoH computations: the deployed target (euler_bms_soh → recovery-aware clean → **hybrid** soh_target) *and* the physically-independent coulomb yardstick (euler_full_charge_soh) → Features → Feature Store → Train euler_model (rate + trajectory + P10/P90 band) → LOV0 backtest → **Gate** (euler_accept_gate scores candidate vs incumbent on the coulomb-confirmed decliner cohort) → **ConditionStep**: register Approved only on PASS → Batch score. Deploys a real model (euler_20260707). This is the only fleet where model-quality alarms are immediately meaningful — it has genuine decliners.

10.2 · Mahindra — two-tier, no gate

Two ingest paths. **Intellicar** (signed current → coulomb SoH → isotonic) covers the ~224 both-feed vehicles; the **native** feed (the 70M-tiny-files problem → *compaction is mandatory*) covers the ~98 % with no sensors, scored by the **Bayesian behaviour model** (bayes_degradation, validated MAE ~1.5 pp). No gate — the coulomb yardstick is too sparse (~18 usable vehicles) and the soh target is already isotonic-clean. Register the forecaster with its honest caveat (it does not beat persistence on the noisy cohort); the behaviour model carries native-only coverage. Monitor: odometer availability + behaviour-model inputs (age, km).

10.3 · Bajaj — reported SoH, young fleet

Dense ingest → SoH = monthly-median of the reported essBmsSohcEstPercValue, kept non-increasing → features **without** current/voltage (charge-cycles, pack temp, ambient temp, drive-efficiency, odometer ÷ 1000) → Train bajaj_model → Backtest (this one **beats** persistence on decliners, 1.40 vs 1.85) → Register → Score. No gate (no current at all → no independent yardstick). Warranty-risk uses the **km-effective deadline** (120k usually binds before 5 years). Monitor: suppress premature drift alarms while the fleet is young.

10.4 · Piaggio — coulomb via intellicar, no gate

Intellicar ingest (**308k tiny files → compaction is the #1 cost lever**) for signed-current coulomb SoH; the native feed supplies thermal/usage features + a distance-per-SoC cross-check (native voltage is 100 % null). Train model → Backtest (does **not** beat persistence, 3.69 vs 2.93 → register but flag low-confidence, communicate with wide bands). No gate — building the yardstick would mean re-scanning all 308k files.

10.5 • Montra — new-fleet placeholder

Day-sampled 10-vehicle extract (montra_sample) → SoH via BMS remaining-capacity (current is **un-signed** → coulomb impossible) → features (has real pack temperature, a plus) → Train model → **flat placeholder** (0 decliners) → register as PendingManualApproval / tagged placeholder → score with a wide caveat. No gate. Monitor: **cold-start — suppress “perfect-fit / no-drift” alarms**; the model becomes real as the fleet ages or more vehicles are onboarded via montra_sample.

10.6 • Onboarding the next OEM

Audit its feed → drop it into the \$10 decision tree to pick soh_method → add a parameter set (model_module, eol_pct, warr_years/km, has_gate) → run the same pipeline. No new pipeline code.

Appendix — script → SageMaker component

Script	Role	Becomes
src/import_cohort.py, montra_sample.py	extract	Processing (or Glue)
src/soh.py, euler_bms_soh.py, euler_full_charge_soh.py	SoH target + independent yardstick	Processing
src/features.py, src/*_features.py	feature engineering	Processing → Feature Store
src/euler_train.py, src/oem_train.py	train + backtest + registry	Training job + Model Registry
src/euler_accept_gate.py	acceptance gate (PASS/FAIL exit)	Processing + ConditionStep
src/data_quality.py	data-thin gating	Processing filter + Model Monitor baseline
src/config.py (CFG, SOH_METHOD, FLEET_WARRANTY)	per-OEM registry	Pipeline parameters
models/<oem>/registry.json,diag post-manifest_registry.pkl dashboard/*.py	post-manifest_registry.pkl consumption	SageMaker Model Registry repoint to batch-scored S3/Redshift (+ optional endpoint)